Plus Rehash Incrementality (Double ML)

Interview preparation pack for Chris - redBus VP of Data Science resume deep-dive

Built only from: project notebooks (`May25_DML.ipynb`, `Feature_importance.ipynb`), project docs 01-05, `Trimming_Result_and_Manager_Narrative.md`, and the two Plus_Incr PDFs. Anything not in those sources is marked **not in files**. No number here is invented.

How to use this pack

1. Read Section 1 first. Your files contradict each other on the headline number; fix that before you rehearse.
2. Learn Section 3 (concepts) until you can say it without notes. A VP will probe the "why", not the code.
3. Rehearse Section 8 (stories) out loud with a timer. Then drill Section 9 (cross-questions).
4. Night before: Section 12 (one-page cheat sheet) and Section 11 (resume wording).

#	Section	#	Section
1	Status: conflicts and open items	8	Spoken stories (30s, 90s, STAR, backbone)
2	The project on one page	9	Cross-question drill (30 questions)
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1. Status: conflicts and open items (read first)

The files describe at least three different "results". A VP who asks "what was the number?" must get one consistent answer. Resolve these before the interview.

Source	What it says	Read it as
`May25_DML.ipynb` (1M accounts, 500K test / 500K control, 15+ net orders)	ATE = **-138.4** (SE 30.0, 95% CI -197.2 to -79.6, p = 4e-6) on `nm_spends_post_12m`. Raw train gap about +2.5K.	Latest executed notebook. Sign flips vs raw gap. 10-14 bucket is empty (population has no one below 15 net orders).
Monthly summary PDF + Trimming doc (S1, non-mobiles, base 17.1M)	GMV lift **21.1%** (May-Apr 2026). Trimmed to common support: about 23%. MatchIt: 25.8%.	The number your story is built around (documented in two files).
S1 working-sheet PDF (first table)	ATE 11,209 / 23,883, lift 219% / 384% / 303.5%, RPC about 65	Looks broken or a different spec (ATE is about the size of test spend). **Do not quote.**
Same PDF (later table)	May-only: ATE 169, lift 6% (CI 5-8%), RPC about 4	Single-month early read. Different window.

Three issues a VP can find in 60 seconds

A. Which run is final? The notebook (-138) cannot produce 21.1% (+3,863 per member). Decide which dataset/spec the 21.1% came from and be able to name it (sample, control construction, window, trimming). I will not guess.
B. Test/control are defined with post-period conditioning. PDF header: test = "Plus 10+ as of 30 Apr 2025 and Plus in every quarter until Apr 2026"; control = "transacted in at least one quarter". Test members had to persist through the outcome window. That is survivorship / post-treatment selection and likely inflates lift. The profile sheet shows control shrinking from 1.2M to 520,377 (-56.6%) after filtering. Know this cold (see Q10 and Section 10).
C. Date logic. Data shows `ref_date_key` = 20260501, while the PDF defines the cohort as of 30 Apr 2025. Pre/post window dates are **not in files**. Know the exact pre window and post window.

Open items table (fill before the interview)

Item	Why it matters	Your answer
Final result and its run (dataset, control, spec)	Headline number	_____
Currency / unit of spend	Say "INR" or "rupees per member" correctly	not in files
Exact pre and post windows, ref date	Defends no-leakage claim	not in files
ATE or ATT reported	"Impact on Plus members" is ATT	Code default = ATE
Share of S1 dropped by [0.1, 0.9] trim; post-trim SMD; counts	Population-change defence	not recorded
Real decision the result fed (budget, AOP, targeting, SC)	The punchline of STAR	not in files
Resume bullets for this project	Wording audit	not in files

2. The project on one page

Question	Answer
Business question	What is the true incremental 12-month spend of Plus, and for whom? Used to judge return on Supercoin spend and where to push the programme.
Why hard	Plus is opt-in. Joiners were already heavier, broader, more active shoppers, so test minus control overstates impact. No randomized holdout exists.
Method	Double ML, interactive regression model (`DoubleMLIRM`): predict each Plus user's no-Plus spend from pre-period behaviour, impact = actual minus counterfactual, corrected with a propensity model, 5-fold cross-fitting.
Data	1M accounts sample (500K Plus, 500K non-Plus), 215 pre-period features, outcome `nm_spends_post_12m` (non-mobile, 12 months, capped at 1st/99th percentile).
Reported result	About 21% GMV lift (S1), about 23% after trimming to common support, 25.8% with MatchIt control. Notebook run: ATE -138 with strong heterogeneity (positive for light buyers, negative for heavy). **Reconcile (Section 1).**
Biggest strength	Counterfactual built per user from rich pre-period profile; diagnosed overlap as the reason S1 undershoots and showed a dose-response-style move toward MatchIt.
Biggest weakness	No placebo, no sensitivity analysis, test defined by persistence, single spec, outcome capped, heavy-user tail unreliable.
Honest one-liner	"Directionally strong, assumption-dependent, and I know exactly which three checks I would add next."

Pipeline in four lines

`Feature_importance.ipynb` (LightGBM importances for outcome and for selection) -> `select_features_cumsum_60.py` (keep features up to 60% cumulative importance, union of both lists, 215 features) -> `May25_DML.ipynb` (clean, 80/20 split, FLAML-tuned XGBoost nuisances, `DoubleMLIRM`, GATE, CATE, balance diagnostics, Excel export).

3. Concepts in depth

3.1 The causal problem

- **Potential outcomes:** each user has Y(1) (spend with Plus) and Y(0) (spend without). We observe only one. The causal effect for a user is Y(1) - Y(0).
- **Selection bias:** E[Y | Plus] - E[Y | non-Plus] = true effect + baseline difference. Plus joiners had higher baseline spend (train means: 5,963 vs 3,460), so the raw gap (+2.5K) is mostly baseline.
- **Fundamental fix:** build the missing Y(0) for each Plus user from non-Plus users who look similar pre-treatment. Control users are **training data for the no-Plus world**, not the comparison group.

3.2 Estimands

Term	Meaning	In this project
ATE	Average effect over everyone in the sample	`DoubleMLIRM` default score. Reported in the notebook (-138).
ATT	Average effect on those who actually got Plus	What "impact on Plus members" means. `score='ATT'`. Confirm which one fed the 21.1%.
GATE	Group average effect	Buckets of `fk_net_orders_pre_12m`: 10-14, 15-19, 20-24, 25+, and 10-19 vs 20+
CATE	Effect as function of covariates	Spline basis on `fk_orders_pre_12m`, joint 95% CI
Overlap-population ATE	Effect for units with real comparables only (Crump et al. 2009)	After trimming propensity to [0.1, 0.9]

3.3 Three nuisance models (all use pre-period X)

Model	Trained on	Predicts	Role
`g0` (regressor)	Non-Plus users	Spend without Plus	Counterfactual for Plus users
`g1` (regressor)	Plus users	Spend with Plus	Counterfactual for non-Plus users (needed for ATE)
`m` (classifier)	Everyone	P(Plus | X), propensity	Weights the correction, trims non-overlap. Never creates a counterfactual and never moves users between groups.

In code: one `ml_g` (XGBRegressor) and one `ml_m` (XGBClassifier) are passed. `DoubleMLIRM` fits `g0` and `g1` by treatment arm internally; both score every user.

3.4 The score and the intuition

ATE per-user score: `g1(X) - g0(X) + T*(Y - g1(X))/m(X) - (1-T)*(Y - g0(X))/(1-m(X))`, averaged over users. First part = model-based effect. Second and third = propensity-weighted residual corrections.

ATT intuition (worked example from your docs): Plus user, `g0` predicts 1,000 without Plus, actual 1,500, first-pass effect +500. But `g0` was trained mostly on ordinary non-Plus users and may be biased for Plus-like people. Check its error on real non-Plus users who resemble Plus users (actual 1,200 vs predicted 1,000 = +200), weighting them by odds p/(1-p). Corrected effect = 500 - 200 = **+300**.

3.5 Why "doubly robust" and "orthogonal"

- **Doubly robust:** estimate is consistent if either the outcome models or the propensity model is good (strictly, with cross-fitting and rates, the product of errors must be small).
- **Orthogonal score:** the estimate is first-order insensitive to errors in `g` and `m`. That is why flexible ML nuisances are allowed and valid CIs still exist.
- **Cross-fitting:** split into 5 folds; each fold is scored by models trained on the other 4. Removes overfitting bias (nobody is scored by a model that saw them).

3.6 Assumptions and how each fails

Assumption	Meaning	Testable?	Failure mode here
Unconfoundedness	Given X, joining Plus is as-good-as-random	No	Hidden intent, life events, promotions seen only by joiners

Assumption	Meaning	Testable?	Failure mode here
Overlap / positivity	0 < P(Plus | X) < 1 for all X	Partly (propensity, SMD)	AUC 0.887: very separable. Heavy users have few non-Plus lookalikes, so `g0` extrapolates
No post-treatment controls	X measured before treatment	Audit	Name-based filter only; test cohort defined by persistence in post window
SUTVA	No spillover, one version of treatment	No	Possible spillover (shared inventory, delivery capacity); benefits may vary by member or over time (not in files)
Consistent outcome window	Post window fully after enrolment for all	Audit	Control observation window may differ (docs open question)

3.7 Why DML and not the alternatives

Method	Weakness for this problem	When I would pick it instead
Linear regression with controls	Misspecified for nonlinear behaviour across 215 features; cannot flex	Few covariates, small data, need simple explanation
Propensity score matching	Discards data, sensitive to PS model and caliper, one estimand; MatchIt gave 25.8% with about 20K controls and duplicate nearest-neighbour matches	Need transparent balance story, small pool
Plain ML (T-learner)	Regularization bias leaks into effect; no valid CI	Pure prediction
IPW alone	High variance with extreme weights	Strong overlap, good propensity model
DiD / synthetic control	Needs a parallel-trend or donor-pool story; needs pre-trend data per user group and an untreated comparison timing	When a policy change hits groups at different dates (listed as next-step triangulation)
RCT / holdout	Not feasible for standing programme	Always the gold standard going forward
Uplift / meta-learners (X, R, DR)	Need a targeting use-case and calibration (Qini) which needs randomization	Targeting decisions with a randomized pilot

3.8 PLR vs IRM

PLR (partially linear regression) assumes a constant effect and a linear treatment term. IRM is built for **binary treatment**, allows heterogeneity, and gives the doubly robust AIPW-style score. Plus is binary and effects clearly vary by order bucket, so IRM.

3.9 Trimming and overlap

- `trimming_threshold=0.01` only clips the extreme 1% of propensities in the score. It does not fix extrapolation.
- Common-support trimming to [0.1, 0.9] removes units with no honest counterfactual. It **changes the estimand** to the overlap population. Say so.
- King and Zeng (2006), "The Dangers of Extreme Counterfactuals": model-based estimates for covariate profiles far from the data depend on functional-form guesses.

4. Setup: what was actually built

4.1 Treatment, outcome, population

Item	Detail	Source
Treatment	`cohort`: test = 1 (Plus), control = 0	Notebook
Outcome	`nm_spends_post_12m`. Switch at top of notebook (alternatives: orders, units). NaN filled with 0.	Notebook
Population	1,000,000 unique accounts; 500,000 test, 500,000 control	Notebook
Eligibility (PDF)	Test: Plus 10+ as of 30 Apr 2025 and Plus in all four quarters (May-Jul, Aug-Oct, Nov-Jan, Feb-Apr) to Apr 2026. Control: 10+ non-Plus, non-VIP, May 2025-Apr 2026, transacted in at least one quarter.	S1 sheet PDF
Net orders support	All accounts have `fk_net_orders_pre_12m` of at least 15 in the notebook sample (10-14 empty)	Notebook counts
Train / eval split	80/20, `random_state=42`. DML fit on train (about 800K, roughly 400K per arm). Test used only to score nuisance models.	Notebook
Full programme base	17.1M Plus members (non-mobiles), used to scale per-member lift	Summary PDF / Trimming doc

4.2 Profile evidence of selection

Metric (profile sheet; run label not in files)	Test	Control (before filter)	Delta	Control (after filter)	Delta
Base	600,000	1,200,000		520,377	-56.64%
Orders / member (12M)	27.1	17.9	+51.4%	19.9	+36.2%
Net GMV / member (12M)	26,840.3	10,662.7	+151.7%	13,224.5	+103.0%
Sessions / member (30D)	80.6	47.9	+68.3%	55.9	+44.2%
Days visited (30D)	16.3	11.9	+37.0%	13.4	+21.6%
Months active (12M)	8.2	6.7	+22.4%	7.2	+13.9%

Notebook means: `fk_net_orders_pre_12m` train mean 22.9 (control) vs 32.4 (Plus). Even after the filter, Plus is far heavier. This is the selection DML must remove.

4.3 Features and leakage control

- Starting table about 1,010 columns. Leakage drops: any column containing `post`, `is_customer`, `lockin`, plus an explicit list (account id, ref date key, reseller flag, supercoin/sc burn, `gmv_sc_*`, `txns_with_sc_*`, `units_with_sc_*`, bupc, vintage, plus_flag, cohort).
- Columns above 80% missing dropped (8 EMI/BNPL days-since columns). Imputation: `txn` columns mean, other numeric 0, categorical mode. `VarianceThreshold(1e-5)`.
- Two LightGBM models for importance: regression on outcome, classification on `cohort` (200 trees, lr 0.05, depth 8, 256 leaves, early stop 20, stratified 90/10; selection AUC about 0.875). Keep features up to 60% cumulative importance in each; **union** (outcome predictors cut variance, treatment predictors capture selection). Result: **215 features**.
- Top treatment-model features: `fk_net_orders_pre_12m`, `months_active_pre_12m`, `gmv_pre_12m`, `rfm_platinum`, `super_category_count_pre_12m`. Top outcome features: `fk_12m_gmv_growth_wt_pt9`, `fk_12m_gmv_growth_wt_pt7`, `fk_orders_gmv_pre_12m`.
- Gotcha: the <= cumsum rule drops the feature that crosses 60%.

Leakage flags to audit

Name-based filtering can miss post-period features. Confirm that `fk_12m_gmv_growth_wt_pt9` and `vintage_7m_12m` are computed up to the reference date. Imputation, 1/99 clipping and importance screening ran on the full data before the split (mild leakage). Clipping applied to numerics including the outcome (per docs).

4.4 Nuisance models and DML configuration

Component	Setting
Library	`doubleml` (`DoubleMLIRM`); `flaml` AutoML; `xgboost`

Component	Setting
Tuning	FLAML, XGBoost only, 1,800 s each. Regressor on RMSE, classifier on ROC AUC. Tuned on pooled train data (outcome model has no treatment input), then configs frozen and reused per arm.
`ml_g` best config	n_estimators 4,301, max_leaves 67, min_child_weight 13.05, lr 0.0095, colsample_bytree 0.364, reg_lambda 0.913
`ml_m` best config	n_estimators 4,977, max_leaves 106, min_child_weight 17.67, lr 0.0177, subsample 0.969, colsample_bytree 0.868
DML	`n_folds=5`, `trimming_threshold=0.01`, `random_state=123` on the models, one repetition, score = ATE
GATE	`.gate(groups=...)` on boolean group matrices
CATE	`patsy bs(x, df=5, degree=2)` on `fk_orders_pre_12m`, `.cate()`, `confint(level=0.95, joint=True, n_rep_boot=2000)`
Balance	IPW from `automl_m.predict_proba` on train (1/p treated, 1/(1-p) control), weighted vs unweighted means and SMD, exported to `weighted_vs_unweighted.csv`
Output	`outputs/model_evaluation.xlsx` (metrics, importances, ATE, GATEs, counts, means)

5. Results (every number from the files)

5.1 Nuisance model quality (notebook)

Model	Split	Metrics
Outcome `ml_l` (regressor)	Test	R2 0.3315, RMSE 5,594.1, MAE 3,215.0
Outcome `ml_l`	Train	R2 0.4414, RMSE 5,113.7, MAE 3,024.2 (train-test R2 gap 0.11)
Treatment `ml_m` (classifier)	Test	AUC 0.8871, accuracy 0.8079, precision 0.8251, recall 0.7813, F1 0.8026
Treatment `ml_m`	Train	AUC 0.9443, accuracy 0.8665, precision 0.8862, recall 0.8410, F1 0.8630

Read: AUC about 0.89 means selection into Plus is highly predictable from X. That confirms self-selection and also signals thin overlap. Outcome R2 of 0.33 is modest, which is why the doubly robust correction matters. The train-test gap on the classifier (0.944 vs 0.887) shows overfitting, which also affects the in-sample IPW balance check.

5.2 Raw vs DML

Quantity	Value
Raw means, full 1M	Control 3,654.9; Plus 6,334.1; gap +2,679
Raw means, train 800K	Control 3,460.0; Plus 5,962.6; gap +2,503
DML ATE (notebook)	**-138.4**, SE 30.0, t -4.61, p 4e-6, 95% CI [-197.2, -79.6]
Reported lift (summary PDF, May-Apr 2026, non-mobiles)	**21.1%**; base 17.1M; GMV per base member 22.1K; incremental GMV per member 3,863; SC cost per member 263; total incremental 66.2B; total SC cost 4.5B; RPC 14.7 (non-mobiles) / about 15

5.3 Heterogeneity (notebook, `fk_net_orders_pre_12m` buckets)

Bucket	GATE	95% CI	p	n test / n control (1M)	Raw mean spend test / control
10-14	0 (empty)	-	-	0 / 0	-
15-19	+235.6	[187.5, 283.8]	9e-22	121,554 / 274,555	3,986 / 2,805
20-24	+7.4	[-85.9, 100.7]	0.876	104,795 / 99,881	4,648 / 3,382
25+	-584.1	[-715.0, -453.3]	2e-18	273,651 / 125,564	8,023 / 5,731
10-19 (2-group)	+235.6	[187.5, 283.8]			
20+ (2-group)	-383.8	[-475.9, -291.6]	3e-16		

Earlier cut on gross `fk_orders_pre_12m`: 10-19 +268.0, 20-29 +53.4 (CI -10.6 to 117.3, not significant), 30-39 -468.8, 40+ -557.8. The final reported set is the `fk_net_orders_pre_12m` buckets.

Consistency check (my arithmetic from your outputs)

Weighting the three GATEs by all-user counts (396,109 / 204,676 / 399,215) gives about -138.3, which matches the ATE (-138.4). So the ATE is the all-user average and the buckets decompose it exactly. The old open question "why ATE differs from GATE" disappears for this sample because nobody is below 15 net orders. If an earlier version used buckets that excluded users, group effects need not average to the ATE.

5.4 CATE curve

Spline on `fk_orders_pre_12m` (x from 15 to 126). Effect starts +307 at 15 orders (joint CI about 11 to 604), is about +278 at 16, +253 at 17, +231 at 18, +213 at 19.5, and falls steadily to about -763 at 126 (joint CI about -2,003 to +477 at the right tail). Shape: **falling**. CI is wide at the tail and includes zero. Spline coefficients: 307.1, -68.9, -140.6, -729.3, -1,190.0, -1,070.1.

Plot caveat: the notebook GATE plot cell uses the first `gate` object (10-19/20-29/30-39/40+ on `fk_orders_pre_12m`), not the final `gate2`. Re-plot before showing.

5.5 Balance (docs, from `weighted_vs_unweighted.csv`)

Feature	SMD before	After IPW
super_category_count_pre_12m	0.72	0.31
category_count_pre_12m	0.71	0.30
months_active_pre_12m	0.70	0.31
non_mle_orders_gmv_pre_12m	0.61	0.27
mrp_pre_12m	0.61	0.26
fk_orders_pre_12m	0.60	0.27
fk_net_orders_pre_12m	0.59	0.27
fk_gmv_pre_12m_lt_1k	0.59	0.27
fk_fk_assured_orders_pre_12m	0.54	0.25
gmv_pre_12m	0.54	0.23

Across 215 covariates: mean SMD 0.235 to 0.101; SMD above 0.25: 87 to 8; above 0.1: 169 to 97. Weighting removed over half the imbalance, not all. Residual imbalance is mostly category breadth and 12-month activity.

5.6 The S1 vs MatchIt story (Trimming doc)

Run	Lift	Notes
S1 original (broad about 500K control, trim 0.01)	21.1%	Default trimming clips only extreme 1% tails
S1 trimmed to propensity [0.1, 0.9]	**about 23%**	One-variable change on S1 alone; same data and learners
MatchIt (matched about 20K control)	25.8%	Reference; duplicate nearest-neighbour matches observed

- **Hypothesis:** S1 control pool has poor overlap with Plus customers, especially heavy ones, so `g0` extrapolates, biasing counterfactuals up and lift down.
- **Evidence:** removing only the units with extreme propensity moved S1 toward MatchIt, in the predicted direction.
- **Not proven:** that 25.8% is exactly right (a 2.8-point gap remains: stricter trim, MatchIt tail noise, or different post-trim populations); that the result is free of estimand change; no placebo yet.
- **Not recorded yet:** share of S1 dropped by the trim, test/control counts after trim, post-trim mean absolute SMD (untrimmed 0.30-0.40), per-bucket ATE after trim.
- **Raw-gap argument:** raw pre-period gaps would predict S1 (control about 50% below Test on spend) shows the **larger** lift. It shows the smaller one. DML does not subtract raw averages; it predicts what a Test-like customer would have spent.

5.7 Programme context (monthly summary PDF; separate from DML)

Cumulative window	May	May-Jun	May-Jul	May-Aug	May-Feb 26	May-Apr 26 (non-mobiles)
%GMV lift (overall)	6.5%	7.1%	15.7%	16.3%	26.9%	21.1%
Base (overall)	19.6M	19.2M	19.0M	18.6M	14.8M	17.1M
RPC (overall)	4.5	5.0	12.6	13.3	21	15

Other programme facts in the PDF: TPC grew from 6.5 to 7.38 (84 bps), 20 bps from the Plus programme; 0.5MM MAC added each month due to Plus; repeat for Plus customers improved 250 bps; 30+ TPC repeat 87% to 89%; 12M SCPC grew 3.8 to 4.02. These are **programme outcomes, not DML estimates**. Do not attribute them to the model.

6. Validation: done vs not done

6.1 Done

Check	Result	How to say it
Covariate balance before/after IPW	Mean SMD 0.235 to 0.101; 87 to 8 above 0.25; 97 still above 0.1	"Weighting removed over half the imbalance; I report the residual."
Nuisance fit	Outcome test R2 0.33; treatment test AUC 0.887	"Modest outcome fit, which is why I rely on the doubly robust correction."
Default propensity trim	0.01	"Only clips extreme tails."
Common-support trim experiment (S1)	21.1% to about 23%	"A controlled test of the overlap hypothesis."
Triangulation with MatchIt	25.8%	"A second observational method, not a benchmark against truth."
GATE vs raw bucket means	Exported for plausibility	"Sanity check on direction and size."
GATE-ATE consistency	Counts-weighted GATEs reproduce ATE (about -138)	"Decomposition is internally consistent."

6.2 Not done (gaps) and the honest line

Gap	Why it matters	Say this
Placebo on a pre-period outcome	Direct test for residual baseline bias; effect should be about 0	"Not run yet. It is the first check I would add."
Sensitivity to unobserved confounding (E-value, partial R2 bounds)	Quantifies how strong a hidden confounder must be	"I did not quantify it. I would bound it before presenting."
Robustness grid (importance cutoff 50/60/80, trim 0.01/0.05, XGB/LightGBM/RF, seeds)	Shows the number is not a spec artefact	"Single specification so far."
Propensity histograms / common-support plots	Visual overlap evidence	"I judged overlap from AUC, SMD and the trimming experiment, not plots."
Repeated cross-fitting (n_rep) with median aggregation	Reduces split noise	"One 5-fold split."
A/B or holdout benchmark; CATE calibration (Qini, bucket lift)	Needs randomization	"No randomized arm exists for Plus. I would add a holdout going forward."
Uncapped / log-spend robustness	Outcome is winsorized at 1/99	"Capped outcome; I would show raw and log as robustness."
Heterogeneity by other dimensions (tenure, category breadth)	Order count may proxy something else	"Only order-frequency buckets so far."
DiD around enrolment date	Independent identification	"Listed as triangulation next step."

6.3 Specific weaknesses in the diagnostics

- The IPW balance check scores propensities **in-sample** (model trained on `train_df`, predicted on `train_df`). Train AUC 0.944 vs test 0.887: overfit propensities look more extreme than they should.
- IPW weights are ATE weights (1/p, 1/(1-p)), not ATT weights (p/(1-p) on controls), while the narrative talks about Plus members.
- FLAML tuned on the same training data later used for DML (mild optimism).
- The first SMD printout in the notebook is a wide matrix with constant rows (each column = one covariate); the exported CSV also had 215 duplicated rows. Save one row per covariate.

7. Decisions, code issues, fix list

7.1 Decisions (what the files support)

Decision area	Supported by files?	Safe wording
Return on Supercoin spend (incremental GMV vs SC cost, RPC)	Yes, as a metric: incremental GMV 66.2B, SC cost 4.5B, RPC about 15 (S1, non-mobiles)	"Sized return on Supercoin spend by segment."
Where to invest / which segments to push	Framing in docs (business question), no recorded decision	"Showed lift is concentrated in lighter buckets; tail uncertain."
AOP / budget / construct changes / targeting rules	**Not in files**	Say "informed" unless you confirm a real change. Tell me what happened.

7.2 Code fix list (from docs)

#	Issue	Fix
1	Imputation, 1/99 clipping, importance screening run on full data before split	sklearn `Pipeline` fit inside training folds
2	Default IRM score = ATE; impact on members is ATT	`score='ATT'` or report both
3	GATE plot uses first `gate` object	Plot `gate2`
4	Variable named `obj_dml_plr_fullsample` but class is IRM	Rename
5	`X = list(set(...))` non-deterministic order	`sorted(...)`
6	Hard-coded personal drive paths	Config file
7	<= cumsum drops the crossing feature	< threshold then add crossing feature
8	Tune on all train then reuse configs in cross-fitting	Acceptable; note it or nest tuning
9	Winsorized outcome	Report raw and log-spend robustness
10	215 duplicate rows in SMD CSV	One row indexed by covariate

7.3 Improvements, ordered by value

- Sensitivity analysis for unobserved confounding (partial R2 / E-value).
- Placebo on a pre-period outcome for S1, trimmed S1 and MatchIt (expect about 0%; non-zero S1 is proof of baseline bias).
- Trim-threshold sweep [0.05, 0.95], [0.1, 0.9], [0.15, 0.85]; monotone rise toward about 25% would be a dose-response argument.
- Robustness grid: cutoff 50/60/80, trim 0.01/0.05, learners, seeds; plot ATE across the grid.
- Propensity histograms and common-support restriction.
- Repeated cross-fitting with median aggregation.
- Per-TPC-bucket GATE on trimmed S1 (prediction: 10-14 moves most, 20-24 barely).
- Redefine cohort by baseline enrolment to remove persistence conditioning.
- Convert CATE into targeting rules with expected incremental spend vs benefit cost.
- Triangulate with DiD around enrolment date; add a holdout going forward.

7.4 Confidentiality

Data, feature names, spend figures and effect sizes are employer-confidential (doc 05). In the interview describe results **directionally** (for example "around a fifth") unless you are cleared to share exact figures.

8. Spoken stories

Punchline first, then decision, then method. Say these aloud; do not read them. Version A assumes the 21.1% S1 run is your headline. If instead the notebook is final, use Version B.

8.1 Version A: 30 seconds

30-second version

"Plus is opt-in, so test versus control massively overstated its value because members were already our best shoppers. I used Double ML to estimate what each Plus member would have spent without Plus, and measured impact as actual minus that counterfactual. The honest answer was about a fifth in GMV lift, far below the raw gap, which we used to judge Supercoin return by segment. [Add the real decision.]"

8.2 Version A: 90 seconds

90-second version

"Leadership needed to know whether Plus actually creates spend or just attracts people who already spend. A clean holdout was not possible, and members were far heavier shoppers, so the raw gap told us almost nothing.
I trained models on non-members to predict spend without Plus from twelve months of pre-period behaviour, applied them to each member, and took actual minus predicted. Because those models are imperfect, a second model of who joins Plus corrects the residual error, and cross-fitting stops the models from memorizing the people they score.
The headline was roughly a fifth in GMV lift. The interesting part was a diagnostic: our broad control pool had very few lookalikes for the heaviest members, so the model was extrapolating. When I trimmed only those units, the number moved up toward a matched-control estimate in the predicted direction. So I presented a range with that caveat, not a single point.
What I would add next is a placebo test, a sensitivity bound for hidden confounders, and a redefinition of the cohort by enrolment date."

8.3 Version B (if the notebook -138 is final): 30 seconds

Alternate 30 seconds

"Plus members spent about 2.5K more than non-members, but that was selection. After Double ML adjustment the average effect was near zero overall, positive for lighter buyers and not distinguishable from zero in the middle, with heavy buyers unreliable because there were few comparable non-members. The takeaway: Plus value is concentrated in lighter segments and measurement for heavy users needs a better control or a holdout."

8.4 STAR (decision as punchline)

Part	Script
Situation	Plus is a large, costly Supercoin-funded programme (SC cost about 4.5B vs incremental GMV about 66B in the S1 view). Leadership wanted its true incremental value. No holdout; self-selected members.
Task	Estimate 12-month incremental spend and show for whom it works, defensibly enough to guide spend.
Action	Built per-member counterfactuals with DML (XGBoost nuisances, 5-fold cross-fit, 215 pre-period features). Diagnosed overlap as the reason one control construction undershoots; trimmed to common support and cross-checked with matching.
Result (punchline)	A defensible lift range (about 21-26%) with an explicit overlap caveat and a return-per-Supercoin view by segment. **[Decision and impact: not in files. Insert the real one.]**
Learning	Overlap, not model choice, was the main driver of uncertainty. Next time: placebo and sensitivity first, cohort defined at enrolment, holdout in place.

8.5 Five-line technical backbone

- **Model:** `DoubleMLIRM`, XGBoost nuisances (FLAML-tuned), 5-fold cross-fitting, 215 pre-period features, ATE (ATT is the member-impact estimand).
- **Why DML:** nonlinear behaviour and many features; valid CIs; doubly robust. Regression is misspecified, PSM discards data, T-learner has regularization bias.
- **Assumptions:** unconfoundedness given pre-period behaviour, overlap, no post-treatment controls, no spillover.
- **Validation:** balance before/after weighting, nuisance fit, trimming experiment, MatchIt triangulation.
- **Limits:** no placebo or sensitivity yet, test defined by persistence, outcome capped, estimand becomes overlap population after trimming.

8.6 Three lines to remember (from your docs)

- **Control = training data for the no-Plus world.**
- **Classifier = weights the correction, never creates the counterfactual.**
- **Cross-fitting = nobody is scored by a model that saw them.**

8.7 Explain it to a non-technical VP (analogy)

"If you want to know what a runner would have clocked without new shoes, you do not compare them to the slowest club runner. You look at runners with the same training history who did not buy the shoes, predict the runner's time from that history, and compare. Then you check how good that prediction is on runners you know the answer for."

9. Cross-question drill (30 questions)

Cover the answer, say yours aloud, then compare. Ordered from the twelve hardest to supporting questions. Traps are in grey.

Q1. Why not a randomized holdout?

Plus is a standing programme, so withholding it from a random slice was not feasible here (confirm the reason you actually had). Given that, observational causal inference was the only route, and I say plainly it is assumption-driven. Going forward I would propose a small holdout or a staggered rollout.

Trap / follow-up: Do not say "RCT was impossible". Say it was not feasible for this programme.

Q2. How do you know there is no unobserved confounding?

I do not know it; it cannot be tested. I control for about 215 pre-period behavioural features covering activity, category breadth, spend growth and recency, which proxy intent well, but I have not yet quantified the risk. I would run an E-value or partial-R2 bound and a pre-period placebo.

Trap / follow-up: Never claim "DML removes all bias". It removes bias from observed confounders only.

Q3. Could any control be post-treatment (bad control)?

I dropped columns containing post, is_customer, lockin, plus an explicit list of Supercoin and Plus-flag columns. That is a name-based filter, so I would audit the top features such as the 12-month GMV growth feature for date logic. Bigger risk is the cohort definition: test required Plus in every quarter.

Trap / follow-up: Know exactly which features end at the reference date.

Q4. Overlap: how bad is it and what did you do?

Propensity AUC is 0.887, so groups are very separable. Heavy members have few non-Plus lookalikes, so the outcome model extrapolates. Default 1% trimming did not fix it. Trimming to [0.1, 0.9] on S1 moved lift from 21.1% to about 23%, in the direction predicted. That changes the estimand to the overlap population, which I state.

Trap / follow-up: Be ready for "what share did you drop?" Answer: not recorded yet; I will report it.

Q5. Why cross-fitting?

Flexible models memorize. If a model scores a user it trained on, its residual is too small and bias leaks into the effect. With cross-fitting each user is scored by models that never saw them, which together with the orthogonal score gives valid inference.

Trap / follow-up: Do not say it prevents overfitting of the nuisance models; it prevents bias from it.

Q6. Why XGBoost nuisances and does R2 of 0.33 undermine it?

Behaviour is nonlinear with many interactions. DML needs the product of outcome and propensity errors to be small, not each perfect, so a modest outcome R2 is tolerable because the propensity part corrects. I tuned with FLAML on the training data, which is mildly optimistic.

Trap / follow-up: Do not say high R2 matters for causal validity.

Q7. ATE or ATT, and which did you report?

The code uses the default ATE. The question "impact on Plus members" is ATT. I would rerun with ATT or report both. [Confirm which one fed the 21.1%.]

Trap / follow-up: If you do not know, say so; do not guess.

Q8. Why is the effect negative for heavy users? How did you validate CATE?

I would not read it causally. Overlap is worst there: in the 25+ bucket test is 274K vs control 126K, and the CATE tail CI includes zero. Most likely extrapolation error. There is no ground truth for CATE and no randomized arm for Qini, so I limit claims to segments with common support and treat the heavy tail as unreliable.

Trap / follow-up: Do not say "heavy users are cannibalized" without evidence.

Q9. S1 says 21% and MatchIt 25.8%. Which do you trust?

I report a range. S1 undershoots because of extrapolation for heavy members; trimming closed about 2 of the 4.7 points. MatchIt has a small noisy pool with duplicate matches. Neither is truth; I would use 21% as conservative and the trimmed value as a better central estimate, with a placebo to adjudicate.

Trap / follow-up: Do not pick the higher number because it looks better.

Q10. Your test group is Plus in every quarter. Is that survivorship?

Yes, it is a real risk. Test members had to persist through the outcome window while control only needed one transacting quarter. That conditions on post-period behaviour and likely inflates lift, direction not certain. Fix: define test by baseline enrolment (intent to treat) and compare, then show the sustained-member effect separately.

Trap / follow-up: This is the single most dangerous question. Own it calmly.

Q11. How was it used?

The result fed a return-per-Supercoin view (incremental GMV vs SC cost) by segment. [Insert the real decision: not in files.]

Trap / follow-up: Do not claim budget or AOP impact you cannot show.

Q12. What would you do differently?

Fit preprocessing and screening inside folds; report ATT; add placebo, sensitivity and a robustness grid; run raw and log-spend robustness; redefine the cohort by enrolment; put a holdout in place for future measurement.

Q13. Explain double robustness in one sentence.

If either my spend model or my who-joins model is reasonable, the estimate is still pointed at the right answer, and the propensity weighting repairs mistakes in the spend model.

Q14. What does the classifier actually do?

Estimates who selects into Plus. Used to weight outcome-model errors and to trim units with no comparable controls. It does not predict spend and never moves anyone between groups.

Trap / follow-up: Do not say it reclassifies users.

Q15. Why two outcome models g0 and g1?

g0 trained on non-Plus predicts spend without Plus; g1 trained on Plus predicts spend with Plus. Both score everyone; the model-based effect is g1 minus g0, and residual corrections are added.

Q16. Why IRM and not PLR?

Binary treatment, heterogeneous effects, doubly robust score. PLR forces a constant linear effect.

Q17. Why drop post-period columns?

Controlling for something Plus itself changes (post-period spend, Supercoin burn, lock-in) blocks part of the effect and biases the estimate (post-treatment bias).

Q18. Why 60% cumulative importance for feature selection?

A heuristic to cut noise and tuning time; taking the union across outcome and treatment lists keeps confounders for either side. I would test 50% and 80% for stability. The rule also drops the feature that crosses 60%.

Trap / follow-up: Do not call it principled; call it practical and untested for sensitivity.

Q19. Why not just compare Plus to non-Plus?

Self-selection. Plus members were already heavier shoppers (for example orders per member 27.1 vs 17.9, GMV per member +152% in the profile sheet), so the raw gap is mostly baseline.

Q20. Why not matching or regression?

Behaviour is nonlinear with many features. Matching discards data and is sensitive to the propensity model; regression is misspecified. DML handles both and gives valid CIs. I did use matching as a cross-check.

Q21. What is orthogonality?

The effect estimate is first-order insensitive to errors in the nuisance models, so ML can be used for them without breaking inference.

Q22. Residual imbalance: 97 features still above 0.1. Is that a problem?

Weighting cut mean SMD from 0.235 to 0.101 but did not eliminate it. The doubly robust outcome model absorbs leftover imbalance. I would tighten it with a better propensity model or an overlap restriction, and I report it.

Trap / follow-up: Admit the in-sample propensity weakness if probed.

Q23. How big are the samples and did you use all 17.1M?

1M-account sample (500K per arm), DML fit on about 800K train. The 17.1M is the programme base used to scale per-member lift to total incremental GMV.

Q24. Why is the outcome capped, and does it matter?

Winsorizing at 1st/99th percentile limits the pull of extreme spenders on tree models and variance, but it also truncates the heavy-user tail where effects may be largest. I would show uncapped and log-spend robustness.

Q25. Could there be spillover (SUTVA)?

Possibly, through shared inventory, pricing or delivery capacity, though per-user spend effects are likely small relative to the programme. If benefits vary by member or over time, "Plus" is not strictly one treatment. I treat it as a limitation.

Q26. How would a placebo test work here?

Run the identical pipeline on a pre-treatment outcome (for example spend in a window before the cohort date, with features cut earlier). The effect should be about zero; a non-zero value is direct evidence of residual baseline bias. Do it on S1, trimmed S1 and MatchIt.

Q27. How would you explain the result to the CFO?

"Out of every 100 units of Plus-member spend, roughly 17 is genuinely incremental; the rest they would have spent anyway. Return per unit of Supercoin cost is about 15 in the broad view, with the caveat that heavy members are measured less reliably." (Check: 3,863 incremental / 22.1K actual GMV per member = 17.5%; the 21.1% lift is relative to the counterfactual of about 18.2K.)

Trap / follow-up: Keep lift (vs counterfactual) and share of spend (vs actual) distinct.

Q28. What if a stakeholder asks you to just report the higher MatchIt number?

I would report the range and why they differ, and explain that the higher number comes from a small, noisy control pool. Picking by outcome is not defensible; I would agree a pre-registered primary spec with the stakeholder before the next run.

Q29. How would you set up a future experiment to measure Plus?

Randomized holdout or staggered rollout at the eligible-user level, primary metric 12-month spend and orders, pre-registered analysis, power for the lightest bucket, and use the holdout to calibrate the DML model (compare DML on the observational population vs the RCT estimate).

Q30. Why did you take on this project and what was your role?

[Your words; not in files.] Frame: leadership needed a defensible incrementality number; you owned design, modelling, diagnostics and communication.

Trap / follow-up: Do not claim team work as solo or vice versa.

10. Honest-gap scripts ("I did not do X, here is how I would")

If asked about	Say	Then add
Placebo / falsification	"I did not run a placebo on this project."	"I would rerun the same pipeline on a pre-period outcome; it should be zero, and I would do it for each control construction."
Sensitivity to hidden confounding	"I did not quantify it."	"I would compute an E-value or partial-R2 bound and report how strong a hidden confounder would need to be to erase the effect."
A/B benchmark	"There is no randomized benchmark for Plus."	"I triangulated with matching and would add a holdout for future calibration."
Qini / uplift calibration	"I did not do uplift calibration; it needs randomization."	"If we targeted on CATE I would first run a randomized pilot and score Qini or bucket lift there."
Test defined by persistence	"That is a limitation of the cohort definition."	"I would redefine by baseline enrolment and compare intent-to-treat."
Robustness	"One specification so far."	"A grid over cutoff, trim, learner and seed is the next step."
ATT	"The code used ATE by default."	"I would report ATT for member impact."
Uncapped outcome	"Outcome is winsorized."	"I would show raw and log spend."

Do not say

- "The classifier moves Plus users into non-Plus."
- "Test was used for testing the counterfactual model." (There is no ground truth for the counterfactual; it is scored, not tested.)
- "DML removes all bias."
- "Plus customers were too extreme for DML to create counterfactuals." (DML always produces one; it extrapolates.)
- "Pre-period metrics have no role in creating counterfactuals." (They are the inputs.)
- "Control customers are lower-spend, so a lower ATE is just correct."
- Any number you cannot trace to a file.

Corrected statements (from the Trimming doc)

Extreme-propensity customers: "For customers with the most extreme Plus-like profiles, there were almost no comparable non-Plus customers in the broad control pool. DML still generated a counterfactual but had to extrapolate rather than measure. That pulled the overall number down to 21%."

Population change: "Yes, I moved to the overlap population on purpose. For dropped units there is no honest non-Plus comparison; any lift reported for them is a model guess. I report the dropped share."

Why DML does not just use raw pre-period gaps: "The ATE compares actual Plus spend to a predicted spend for Plus-like customers. It is only reliable where real controls exist nearby."

11. Resume claims: defend or reword

Your resume is **not in the project files**, so these are the claim types most likely to be on it. Paste your actual bullets and I will audit line by line.

Likely claim	Risk	Safe wording
"Measured true incrementality of Plus"	Only observed confounders; no placebo or sensitivity	"Estimated incremental lift of Plus using Double ML, triangulated with matched control"
"Drove Supercoin / AOP / budget decisions"	Not in files	"Informed return-on-Supercoin sizing" (until you confirm a real decision)
"Validated against an A/B test"	No A/B exists	"Cross-checked against a matched-control estimate"
"Identified high-lift segments"	Heavy tail negative/unreliable; middle bucket insignificant	"Showed lift concentrated in lighter order buckets; tail uncertain"
"21% lift" as a point estimate	Single spec, overlap-sensitive, capped outcome, persistence cohort	"Roughly 21-26% depending on control construction"
"Causal ML / uplift modelling"	No Qini or randomized calibration	"Causal estimation with Double ML; heterogeneity via GATE/CATE"
"Built production pipeline"	Notebooks with hard-coded paths	"Built an analysis pipeline (feature selection, tuning, estimation, diagnostics)"
Role dates	Timeline: Senior BA Jan 2021-2024; Lead since Jan 2025. Never use 2022.	Use exactly these dates

12. One-page cheat sheet

Topic	Memorize
Problem	Plus opt-in. Raw gap +2.5K (train) is mostly selection. Plus orders/member 27.1 vs 17.9.
Idea	Counterfactual per Plus user from pre-period profile. Control = training data for no-Plus world.
Model	`DoubleMLIRM`; XGB (FLAML) nuisances; 5-fold cross-fit; trim 0.01; 215 features; ATE default.
Data	1M sample (500K/500K), train 800K; outcome `nm_spends_post_12m`, capped 1/99; programme base 17.1M.
Selection signal	Treatment AUC 0.887 test (0.944 train). Outcome R2 0.33 test (0.44 train).
Balance	Mean SMD 0.235 to 0.101; >0.25: 87 to 8; >0.1: 169 to 97.
Notebook result	ATE -138.4, CI [-197.2, -79.6]. GATE: 15-19 +236; 20-24 +7 (ns); 25+ -584. 10-14 empty.
Reported result	21.1% GMV lift (S1). Trimmed about 23%. MatchIt 25.8%. Incr GMV/member 3,863; SC cost/member 263; RPC about 15.
Diagnosis	S1 undershoots because heavy members lack comparable controls; trimming [0.1, 0.9] moved it toward MatchIt.
Top 3 weaknesses	Persistence-defined test cohort; no placebo/sensitivity; overlap/extrapolation in heavy tail.
Next steps	Placebo; E-value; robustness grid; ATT; cohort by enrolment; holdout.
Three lines	Control = training data. Classifier weights, never creates. Cross-fitting = nobody scored by a model that saw them.
Confidentiality	Say "around a fifth" unless cleared for exact figures.

13. Glossary

Term	Meaning
Nuisance model	A model you need (outcome or propensity) but whose coefficients you do not care about
Propensity score	P(treated | X)
Counterfactual	Unobserved outcome under the other treatment
SMD	Standardized mean difference; below 0.1 is balanced (diagnostic, not a trimming rule)
IPW	Inverse propensity weighting
AIPW / doubly robust	Combines outcome model and propensity weighting; right if either is right
Orthogonal score	Estimating equation insensitive to first-order nuisance error
Cross-fitting	Fit nuisances on other folds, score the held-out fold
Positivity / overlap / common support	Every covariate profile in treated has real comparables in control
Extrapolation bias	Error from predicting where the training data does not exist
Trimming (Crump et al. 2009)	Drop units with propensity outside [0.1, 0.9]; estimand becomes overlap population
GATE / CATE / BLP	Group ATE; conditional ATE; best linear predictor of the effect on a basis (spline here)
TPC	Transactions per customer (order frequency), used as the segmentation axis
SC / SCPC / RPC	Supercoin / Supercoin per customer / return per Supercoin cost (incremental GMV over SC cost)
MAC	Monthly active customers
S1 / MatchIt	Broad about 500K control construction / matched about 20K control construction
FLAML	AutoML library used to tune XGBoost nuisances
Winsorizing	Capping a variable at percentile limits (here 1st and 99th)

References you can name

- Chernozhukov et al. (2018), Double/Debiased Machine Learning for Treatment and Structural Parameters.
- Crump, Hotz, Imbens, Mitnik (2009), Dealing with limited overlap in estimation of average treatment effects.
- King and Zeng (2006), The Dangers of Extreme Counterfactuals.
- Bach et al., DoubleML package (Python).

14. Questions only you can answer (fill in before the interview)

#	Question	Your answer
1	Which run produced 21.1%, and is the notebook (-138) a different dataset or the newest rerun? What is the notebook's lift in %?	_____
2	Was "Plus in every quarter" used in the modelled cohort? What are the exact ref date, pre window and post window?	_____
3	What decision, budget or AOP change did the lift actually feed, and is there a number?	_____
4	Which estimand fed the 21.1% (ATE or ATT)?	_____
5	What share of S1 was dropped by the [0.1, 0.9] trim, and what were the post-trim counts and SMD?	_____
6	What is the spend currency/unit?	_____
7	What was your exact role (design, build, communicate, present)? Who consumed the result?	_____
8	Paste your resume bullets for this project.	_____

Send me these and I will tighten the stories, replace the bracketed placeholders, and run a timed VP mock (one question at a time, escalating).